

Kal-El update

Nvidia's Tegra 3 processor. Kal-El will have a companion core to the four cores. It works on low power on simple tasks.

Game swapping

Playr2.com is a web site where you can list the games you want to swap for some other game from the community. It supports PS3, Xbox, Wii, PC, etc

Graphics Cards Test

HOW WE TESTED

Graphics cards are all about the performance and what matters is the frame-rates you're getting for your games. Keeping this in mind, we laid higher stress on the performance parameter than the features. We decided to break up the cards into price brackets so as to give each card a fair chance in its price segment. But first let's have a look at our testing rig.

Test rig

Processor: Intel Core i 7 980X Extreme Edition @ 2.93GHz

Motherboard: ASUS Sabertooth X58

RAM: 2 x 2GB Kingston DDR3 RAM 1333MHz @ 9-9-9-24 timings

HDD: Western Digital Velociraptor 600GB

Operating System: Windows 7 Ultimate 64-bit

Monitor: Alienware OptX 2210 (1920x1080 pixel resolution)

Power Supply: Corsair HX 620W

AMD Drivers: Catalyst 11.8

NVIDIA Drivers: 280.26 WHQL

We installed a clean copy of Windows 7 Ultimate on the 600GB drive and set it to Best Appearance. We installed the synthetic benchmarks such as Unigine Heaven 2.5 and 3D Mark 11. Unigine Heaven 2.5 runs on the Unigine gaming engine which lays stress on tessellation and simulates changing light conditions along with supporting DX 11.

We have introduced the 3D Mark 11 benchmark in place of 3D Mark Vantage, as 3D Mark 11 has a suite of tests which stressed the new features in DirectX 11 including tessellation, compute shaders and multi-threading.

The games used for testing were *Crysis Warhead*, *Crysis 2*, *FarCry 2*, *Resident Evil 5*, *STALKER: Call of Pripyat* and *Dirt 3*. No card review is complete without *Crysis Warhead* as it's very taxing on the GPU and built with scaling in mind. We also benchmarked *Crysis 2* which is the latest addition to the *Crysis* franchise as this game is garnering a lot of buzz around it.

STALKER: Call of Pripyat benchmark has really good lighting and shaders. Plus you can enable DX11 and tessellation. *FarCry 2* is another game that is built with scaling in mind but compared to *Crysis Warhead*, is less taxing on the GPU. *Resident Evil 5* is a third-person shooter with some good physics. *Dirt 3* is a racing simulation game which will be a part of our graphics cards test from now on.

With these games all the major gaming genres have been covered. This will give you a good idea on how to judge the prowess of each card.

The temperatures of the card was also noted when the GPU Utilisation was 100 per cent. MSI Afterburner utility was used to measure the temperature, which was measured after running one instance of Unigine Heaven 2.5.

face width and 960 stream processors as compared to the HD 6770 and HD 6790 which have 800 of them. Similarly, each GTX 465 has 352 CUDA cores as compared to 192 of the

comparitively higher specced cards in this segment.

So let's concentrate on the remaining cards as that's where the fight is fierce. PowerColor HD 6790's 3D Mark 11 score is more than the GTX 550Ti's but in all the gaming benchmarks, the ZOTAC GTX 550Ti AMP Edition just edges out to emerge as the one with the highest frame-rates. But when compared with the EVGA GTX 550Ti and MSI N550GTX Ti, the difference is not more than five frames. PowerColor HD 6790 takes a hit in *Dirt 3* and *FarCry 2*

where it is 10 FPS below the fastest GTX 550Ti.

As always the ZOTAC GTX 550Ti AMP Edition tends to heat up the most, thanks to the highest factory overclock amongst comparable cards. MSI N550GTX Ti, on the other hand, left us quite impressed with load temperature around 50 degrees as compared to ZOTAC AMP Edition card which touches almost 80 degrees. Even the PowerColor HD 6790 was 20 degrees cooler. The MSI R6850 was under 60 degrees on load, despite being factory overclocked.



ASUS ENGTX 560

GTX 550Ti's. As far as performance goes, it's a no-brainer, but the depreciation has got these



Galaxy GTX 560

From ₹5K to ₹7K

This is the category with the maximum number of cards and it's one of the most sought-after price bracket. These cards will give you decent frame-rates for resolutions upto 1680x1050. These cards were also bench-

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